

Actors

1. Users:

- Create and edit profile
- Swipe left/right on other users
- Match with users
- Join hobby-based group chats
- Share availability and interests
- View nearby users (location-based)
- Report or block other users

2. Group Moderator:

- Create hobby or interest-based groups
- Approve or remove members from group chats
- Moderate conversations (remove inappropriate messages)
- Schedule group events or hangouts
- Pin announcements in group chats

3. Administrators:

- Approve or reject account registrations (ISU email validation)
- Manage reported users and safety violations
- Disable or suspend accounts
- Manage app-wide settings
- Monitor usage analytics

Non-Functional Requirements

- Minimum of 100 active users
- The user should not wait longer than 0.5 seconds for any UI response
- The user should not have to wait longer than 2 seconds to sign in
- The next profile upon swiping should be visible instantly/0.5 seconds
- The user should not have to wait longer than 10 seconds for a message to send
- Group moderators should be accepting or denying requests and never leaving them as is
- Location-based recommendations should load efficiently without blocking the user interface.
- Only users with valid Iowa State email addresses can create accounts.
- User location data should only be used for matching and recommendations, not publicly displayed.

Tables and Users

USERS TABLE: Tracks information related to user's account

- User ID - Primary Key
- Name - Full name of the user
- Email - ISU email address
- Password Hash - Encrypted Password
- Bio - About section / user description
- Hobbies - Keywords representing user interests
- Role - Enum(Regular, Moderator, Admin)
- Latitude - Current location(latitude coordinate)
- Longitude - Current location(longitude coordinate)
- Created At - Timestamp of account creation
- Is Active - Boolean flag for suspension status

GROUPS TABLE

- group_id (PK) - ID of the group
- name - name of the group
- description - group description
- created_by (FK = users.user_id) - mod who created the group
- created_at - time group was created

GROUP_MEMBERS TABLE: Tracks user participation within groups

- Membership ID – Primary Key
- User ID – Foreign Key = Users (User ID of member)
- Group ID – Foreign Key = Groups (Group the user belongs to)
- Joined At – Timestamp when the user joined the group

- Status – Enum (Pending, Approved, Removed, Banned)
- Is Moderator – Boolean flag indicating moderator privileges

MATCHES TABLE: Tracks mutual matches between users

- Match ID – Primary Key
- User1 ID – Foreign Key = Users (First matched user)
- User2 ID – Foreign Key = Users (Second matched user)
- Status – Enum (Pending, Matched, Blocked, Unmatched)
- Created At – Timestamp when match was created

REPORTS TABLE: Tracks reports made against users for safety moderation

- Report ID – Primary Key
- Reporter ID – Foreign Key = Users (User who filed the report)
- Reported ID – Foreign Key = Users (User being reported)
- Reason – Description explaining why the report was filed
- Status – Enum (In Review, Approved, Declined)
- Created At – Timestamp when report was submitted

EVENTS TABLE: Tracks group-based events or hangouts

- Event ID – Primary Key
- Group ID – Foreign Key → Groups (Group hosting the event)
- Created By – Foreign Key → Users (User who created the event)
- Title – Name of the event
- Description – Details about the event
- Location – Event location

- Event Time – Scheduled time of event
- Created At – Timestamp when event was created

BLOCKS TABLE: Tracks users blocking other users

- Block ID – Primary Key
- Blocker ID – Foreign Key → Users (User performing the block)
- Blocked ID – Foreign Key → Users (User being blocked)
- Blocked At – Timestamp when the block occurred

CONVERSATIONS TABLE: Tracks chat threads between users or groups

- Conversation ID – Primary Key
- Type – Enum (Direct Message, Group Chat)
- Created By – Foreign Key = Users (User ID of conversation creator)
- Group ID – Foreign Key = Groups (Null, used if conversation is a group chat)
- Match ID – Foreign Key = Matches (Null, used if conversation is between matched users)
- Created At – Timestamp of conversation creation
- Is Active – Boolean flag indicating if conversation is active

CONVERSATION MEMBERS TABLE: Tracks users participating in a conversation

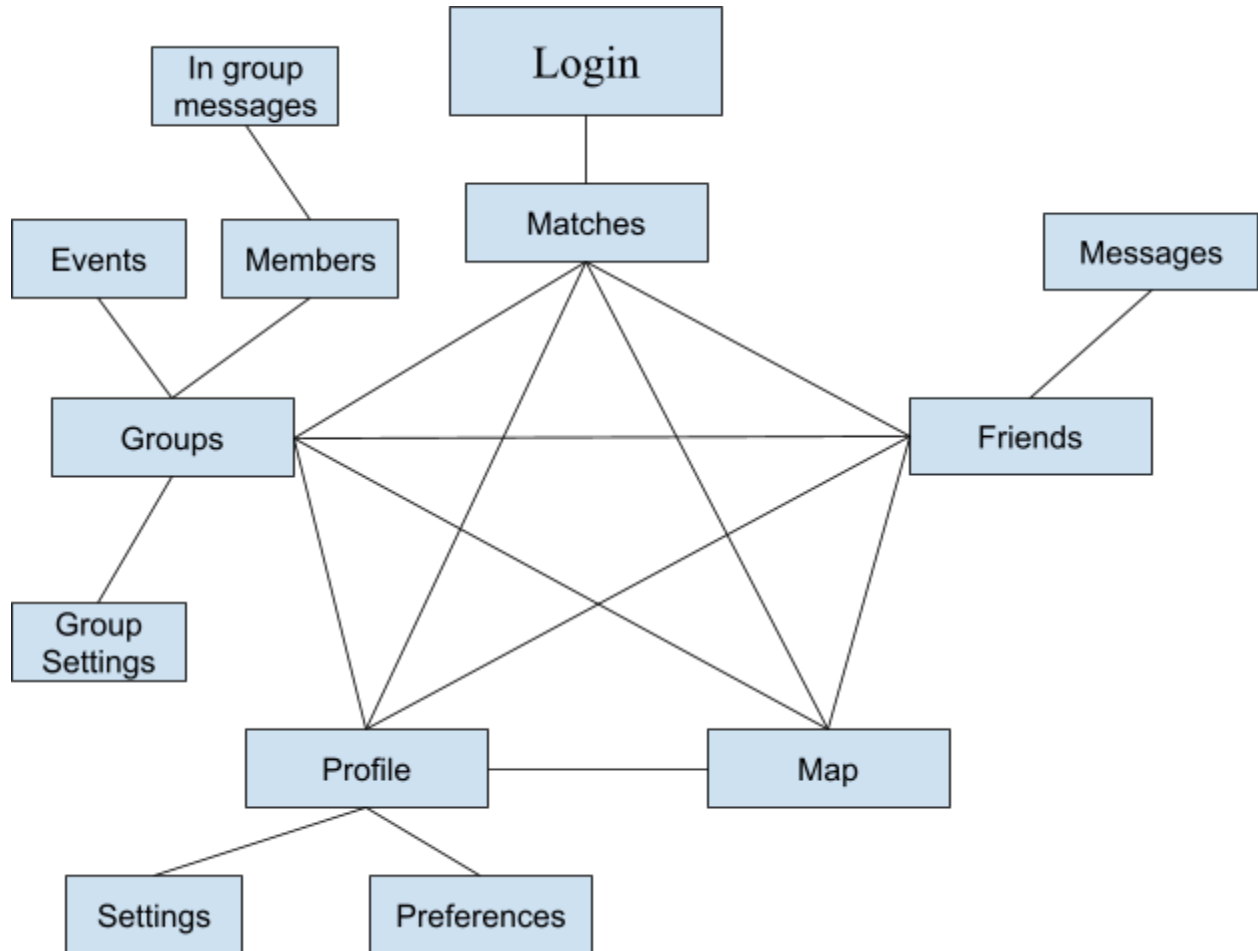
- Membership ID – Primary Key
- Conversation ID – Foreign Key = Conversations
- User ID – Foreign Key = Users

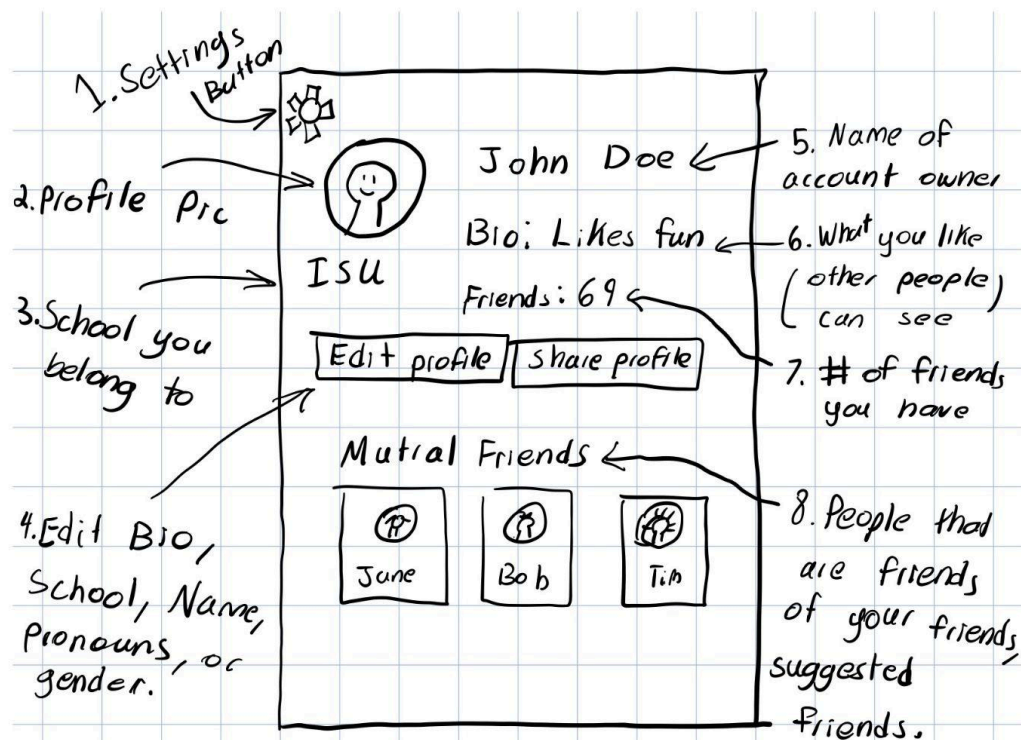
- Joined At – Timestamp when user joined conversation
- Role – Enum (Member, Moderator)
- Status – Enum (Active, Left, Removed, Muted)

MESSAGES TABLE: Tracks individual messages within a conversation

- Message ID – Primary Key
- Conversation ID – Foreign Key = Conversations
- Sender User ID – Foreign Key =Users
- Content – Text body of the message
- Sent At – Timestamp when message was sent
- Edited At – Timestamp if message was edited
- Is Deleted – Boolean flag indicating soft deletion status

Screen Flow Diagram



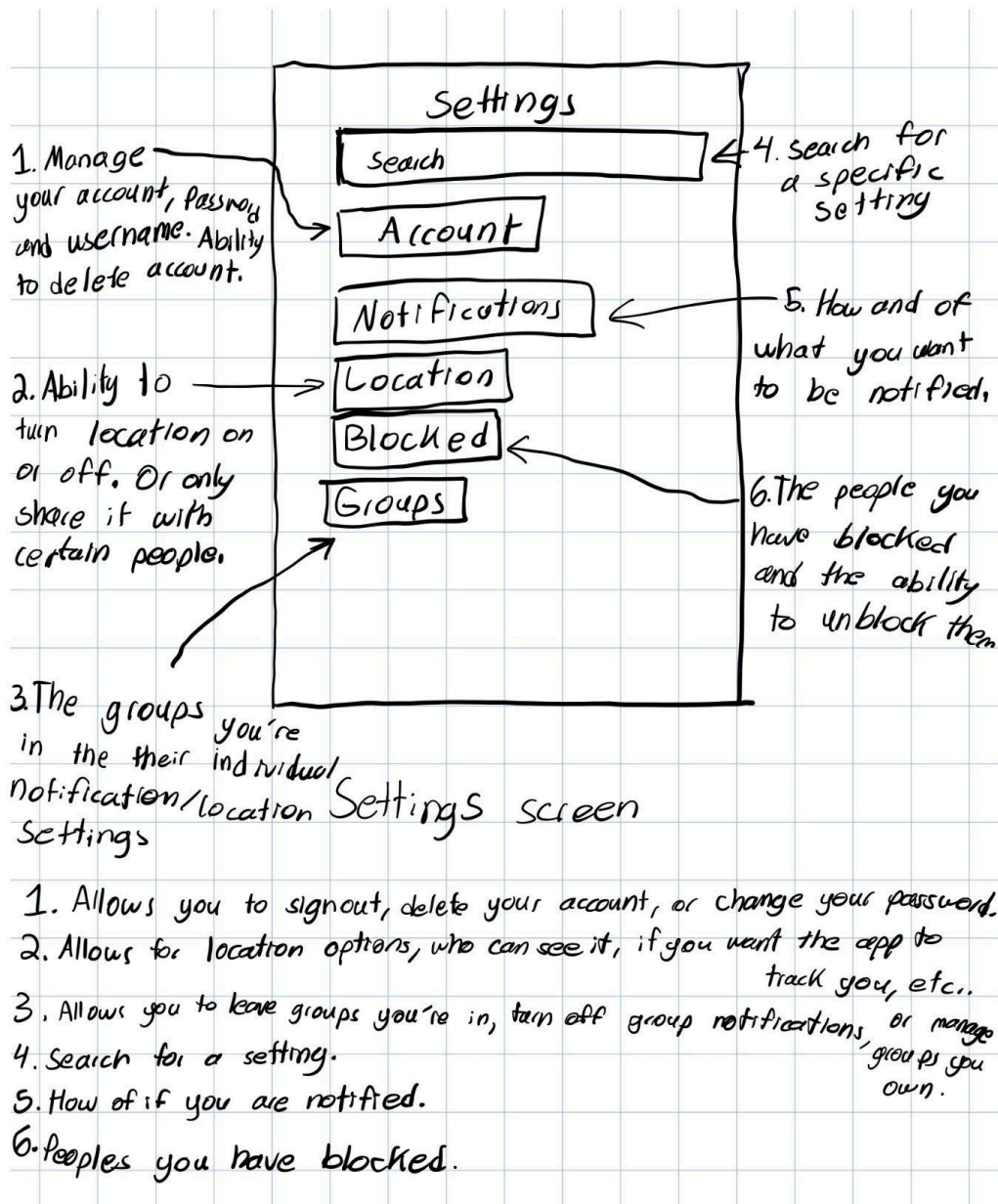


Profile screen

1. A button to bring you to the settings screen.
2. Your public profile picture.
3. Your school (Public info).
4. Place to edit your bio, school, pronouns, profile, etc.
5. Public account name.
6. Things you like, this is also public.
7. How many friends you have.
8. Suggested friends based on current ones.

Jon Buschko

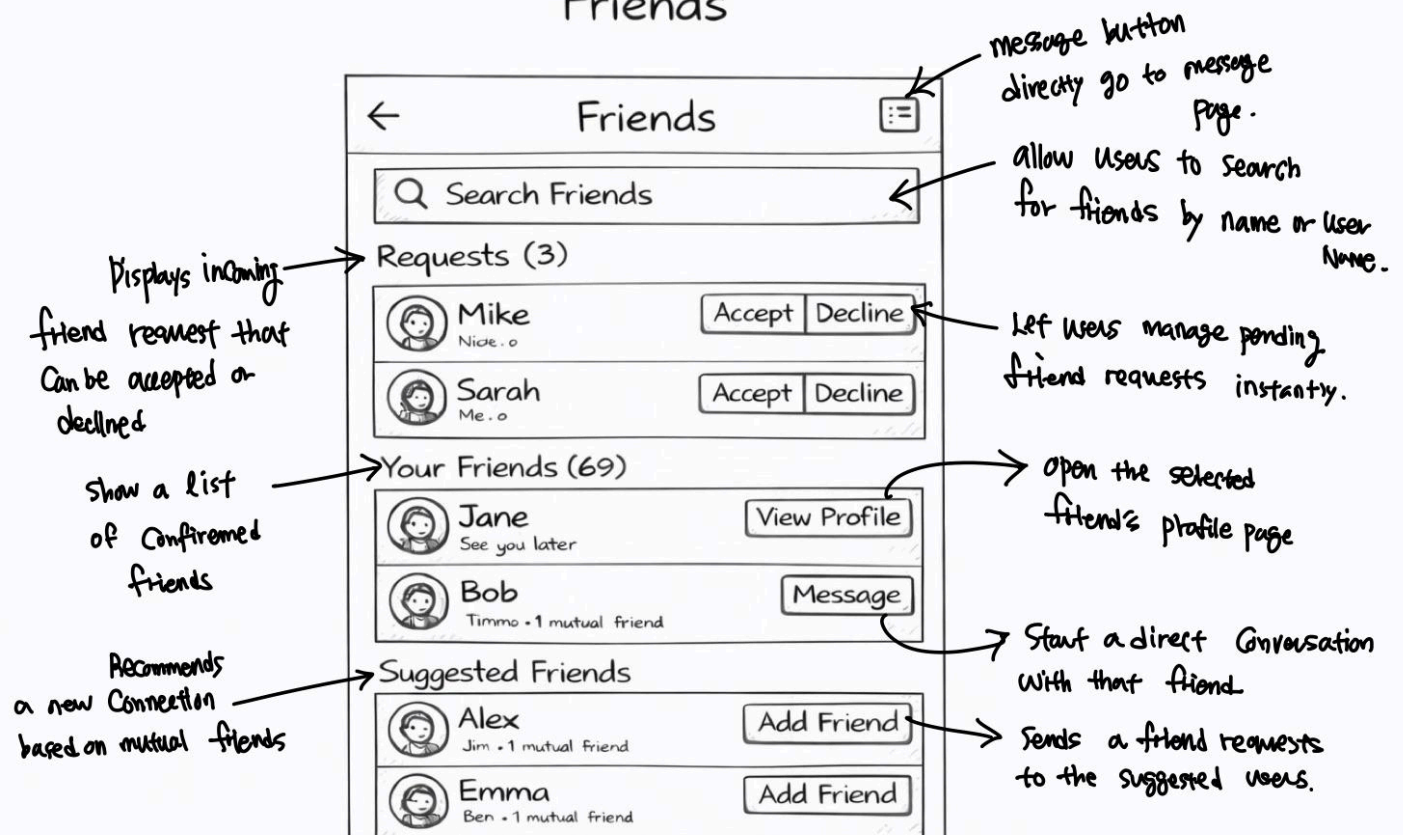
The purpose of this screen is to review, and edit your profile. It allows you to see what your profile looks like to other people.



Jon Buschko

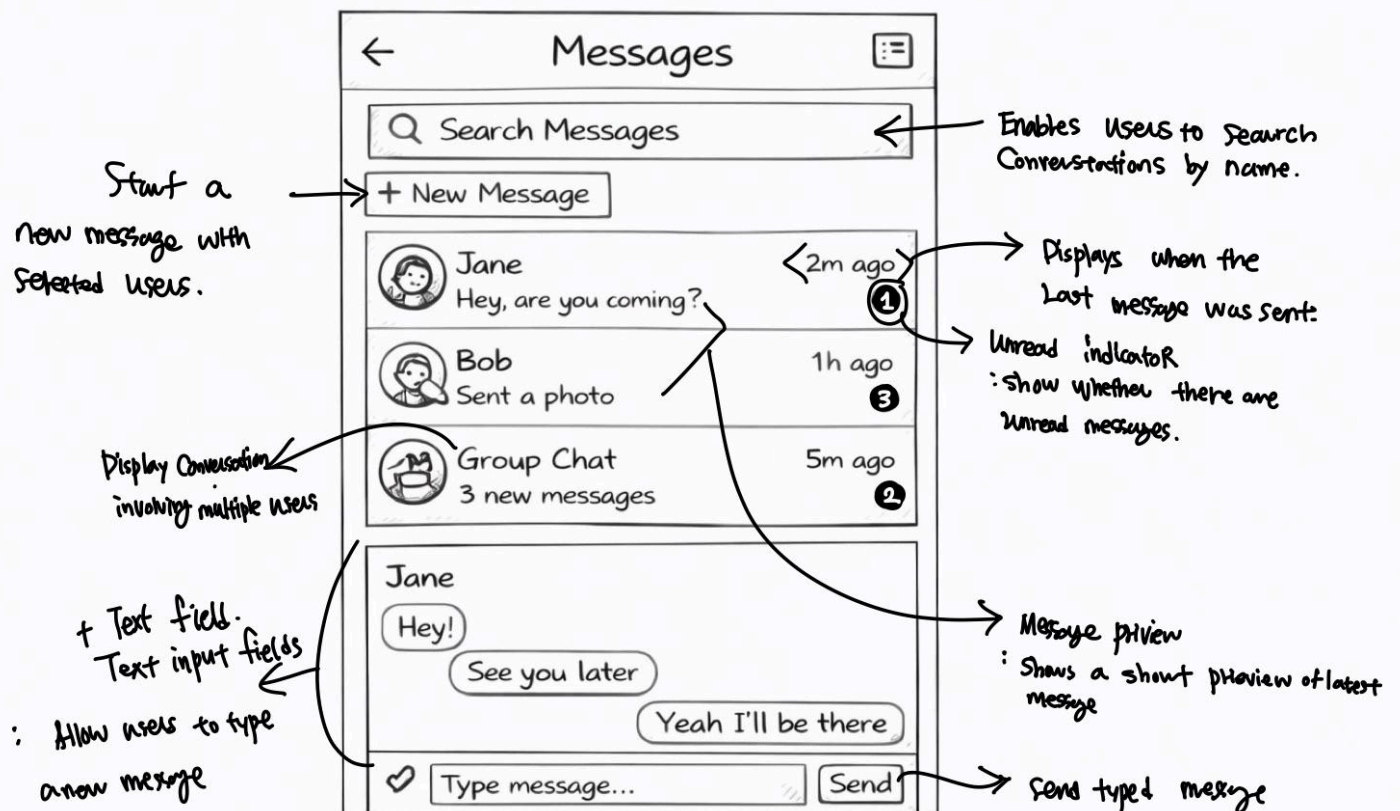
The purpose of this screen is to give you full access to data in and around our app. Including notifications, account details, location, or individual group settings. The default settings will be set upon the user creating their account, the app will ask to track them and if they can give notifications.

Friends



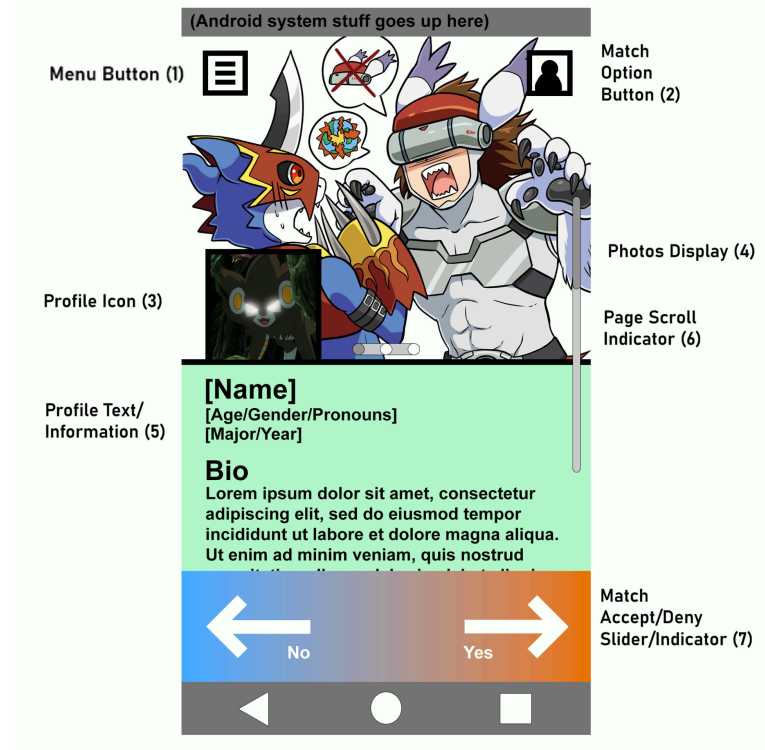
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Messages



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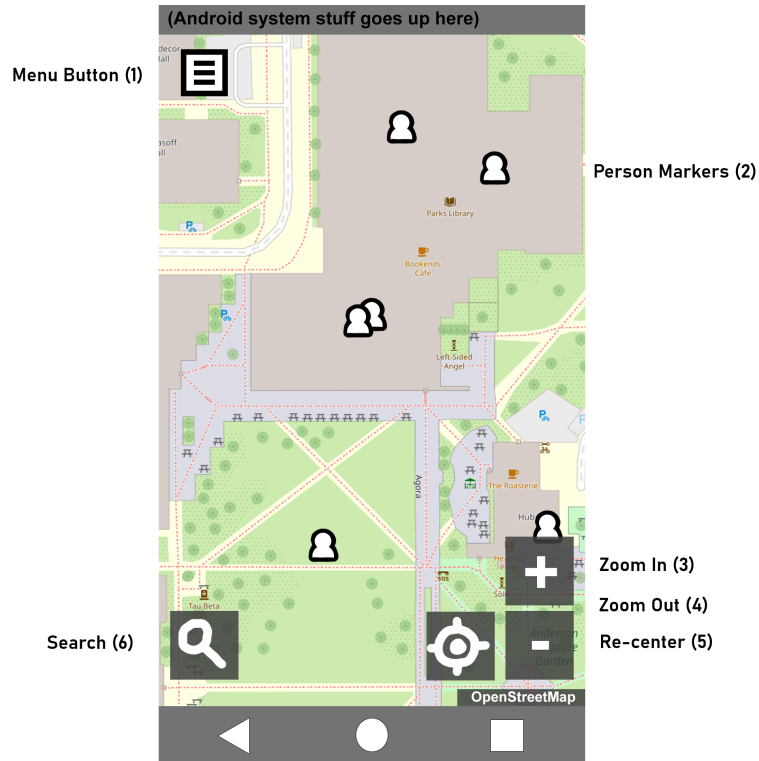
Brendan Petersen - Matches Page



1. Button to open the menu to navigate to other pages (map, messages, etc.)
2. Button to open settings that pertain to the person you have matched with (report user, block user, etc.)
3. The profile icon of the person you have matched with.
4. A display of photos that the person you have matched with has uploaded and chosen to display on their profile
5. Information taken from profile, such as name, age, major, bio, and hobbies.
6. Scroll indicator that shows how far down the page you have scrolled. Used when the profile text requires more space than the phone display allows.
7. Indicator at the bottom of the screen

that shows which way to swipe to approve or deny a match. Important for users who are not familiar with the left/right swiping system.

The purpose of this screen is to implement the matching system that the app relies on. This screen allows users to quickly see other users of the app, and presents information that they can use to choose whether they would like to try matching with this person. On the other side, this page gives users a place to express themselves and their interests to others. This page also provides a place to perform essential functions such as blocking or reporting users.

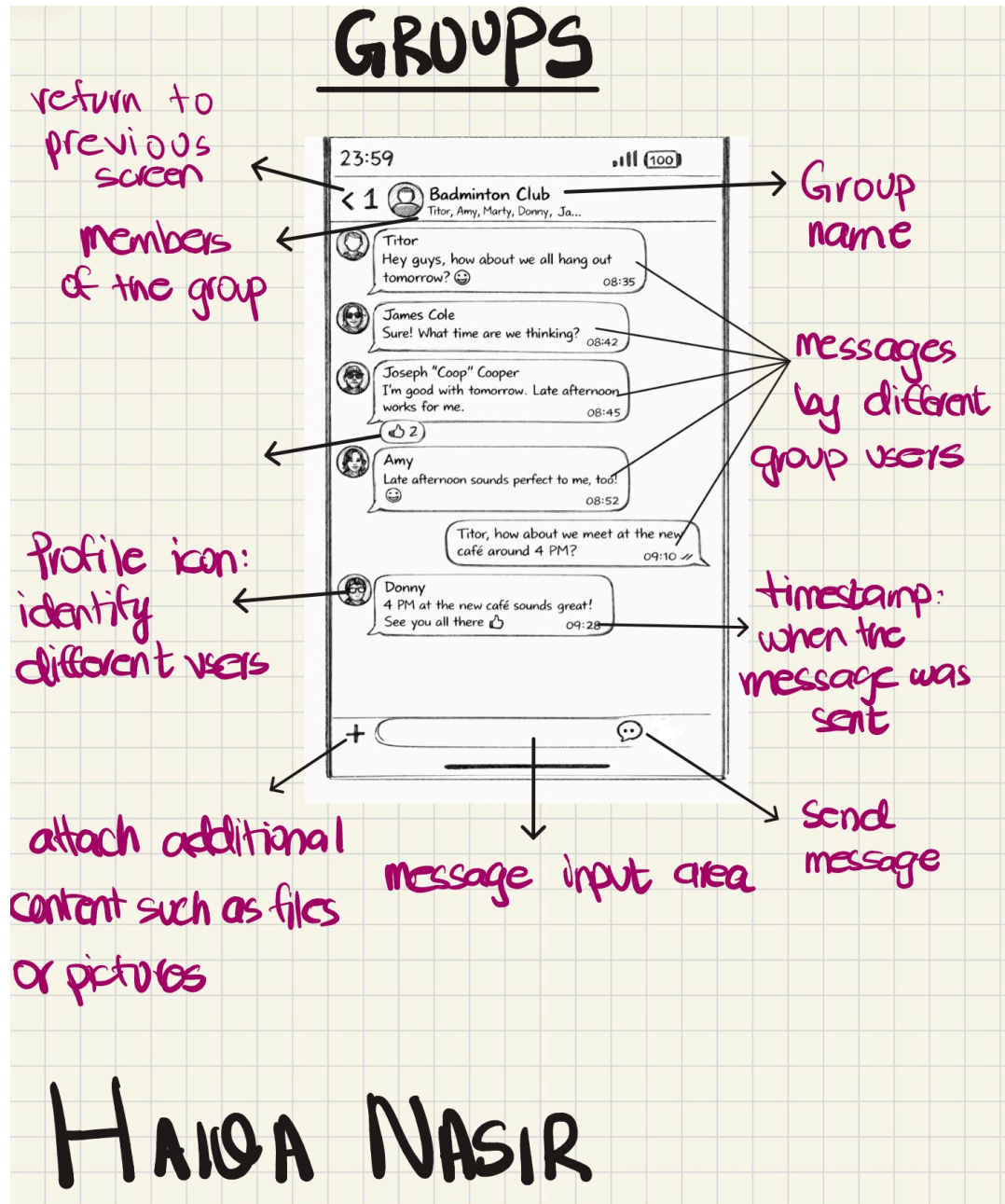


Brendan Petersen - Map Page

1. Button to open the menu to navigate to other pages (map, messages, etc.).
2. Indicators on the map that mark the location of nearby people who are also using the app.
3. Button to zoom into the map.
4. Button to zoom out from the map.
5. Button to re-center the map onto your current location.
6. Button to open a search bar, which can be used to search for locations on the map.

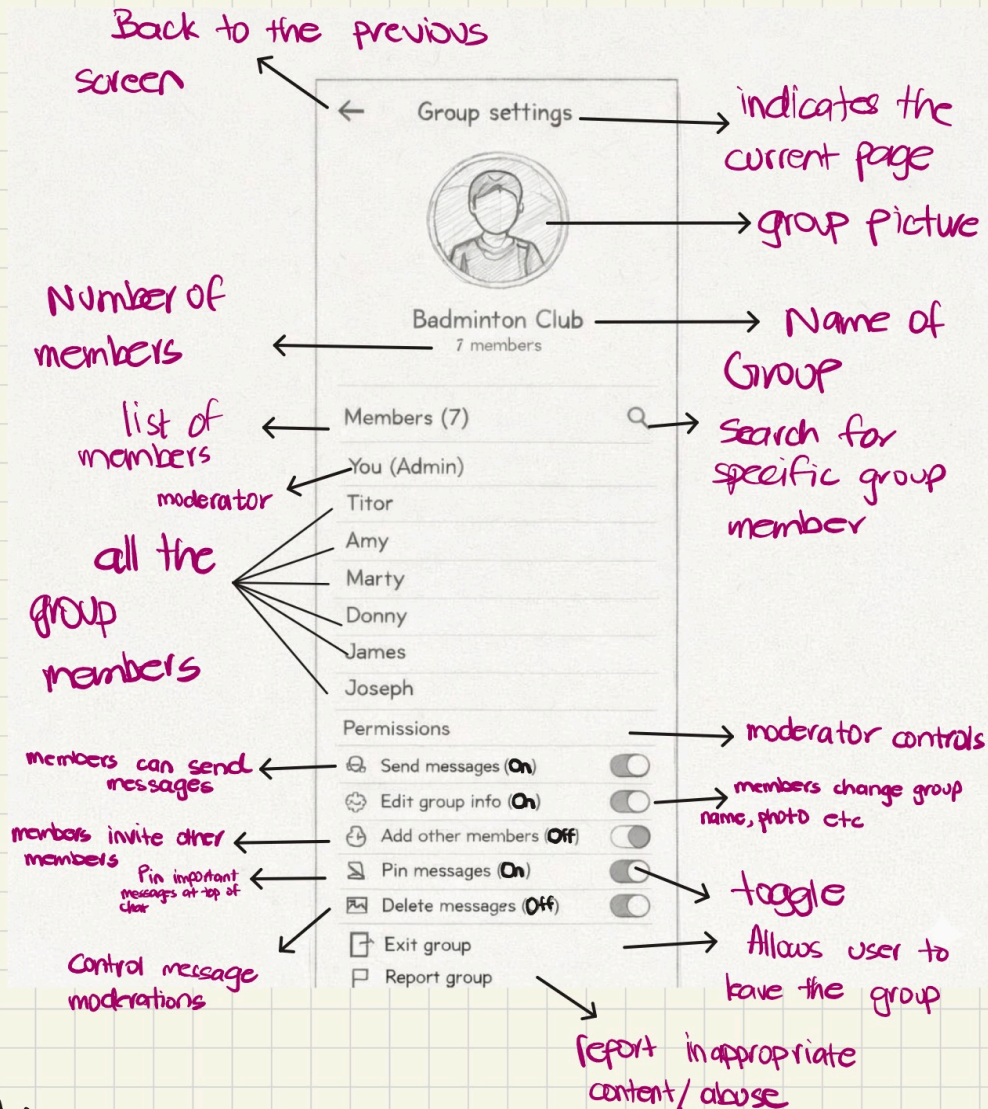
The purpose of this screen is to show where other users of the app are located. Since finding people nearby is a feature of the app, users can use this page to find areas where they could meet more people who are looking to meet others, either through the app or otherwise. This page would also work as a functional map of the campus, though this isn't its primary purpose.

(The map buttons will largely depend on what map service API is used. For this example, I based it off the interface of OpenStreetMap)



The purpose of this screen is to give members of a created group like here the badminton group, a simple, shared space to talk and make plans together. It's where everyone can suggest ideas, discuss timing, and agree on details without things getting confusing. By showing messages in order, along with names and timestamps, the screen makes it easy to follow the conversation and see who said what. The layout keeps the focus on the discussion so users can quickly jump in, respond, and move the plan forward.

GROUP Settings



HANOA NASIR

This group settings screen is designed to give a user/moderator full control over managing and organizing the group. Its purpose is to allow the moderator to oversee members, monitor group size, and adjust permissions such as who can send messages, edit group information, add members, pin messages, or delete messages. It centralizes administrative tools in one place so the moderator can maintain order, enforce rules, manage participation, and ensure the group runs smoothly while still providing options to exit or report the group if necessary.